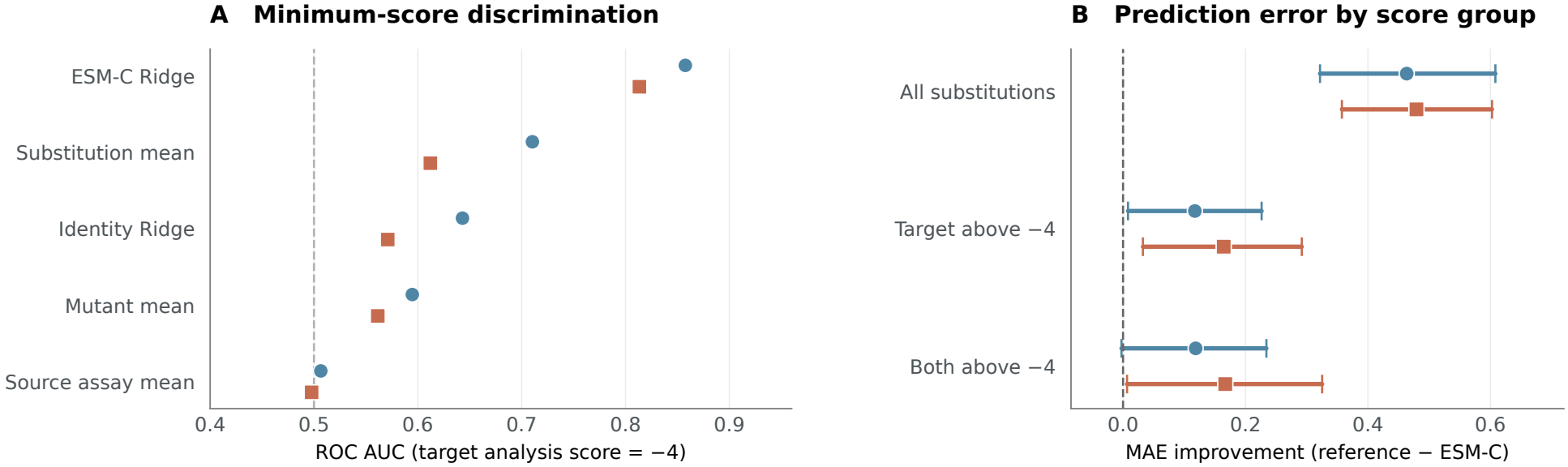
A Minimum-score discrimination
ESM-C Ridge
Substitution mean
Identity Ridge
Mutant mean
Source assay mean
0.4
0.5
0.6
0.7
0.8
0.9
ROC AUC (target analysis score = −4)
B Prediction error by score group
All substitutions
Target above −4
Both above −4
0.0
0.2
0.4
0.6
MAE improvement (reference − ESM-C)
NDM-1 → VIM-2
VIM-2 → NDM-1